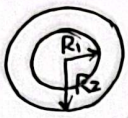
13.2



$$U_{12}=\int_{R_1}^{R_2}E\,dr=\int_{R_1}^{R_2}\frac{q\,dr}{4\pi\varepsilon_0 r^2}$$

在r处的电场 $=\frac{q}{4\pi\varepsilon_0}\left(\frac{1}{R_1}-\frac{1}{R_2}\right)$

$$E=\frac{q}{4\pi\varepsilon_0 r^2}=\frac{U_{12}R_1R_2}{r^2(R_2-R_1)}$$ 方向沿径向向外

13.3 (1) $\varphi_1=\frac{1}{4\pi\varepsilon_0}\left(\frac{q_1}{R_1}+\frac{q_2}{R_2}\right)$

$\varphi_2=\frac{1}{4\pi\varepsilon_0}\cdot\frac{q_1+q_2}{R_2}$

解：$\begin{cases} q_1=6.67\times10^{-10}C \\ q_2=-1.33\times10^{-9}C \end{cases}$

(2) 设距圆心r处电势为零

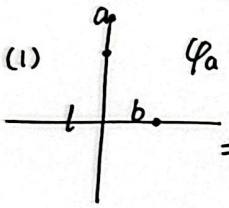
易知 $R_1<r<R_2$

由电势为零有

$$\frac{1}{4\pi\varepsilon_0}\left(\frac{q_1}{r}+\frac{q_2}{R_2}\right)=0$$

解 $r=\frac{R_2}{2}=10\ cm$

13.6 (1)



$$\varphi_a=\int_a^{a+l}\frac{dq}{4\pi\varepsilon_0 r}$$

$$=\int_a^{a+l}\frac{\lambda\,dr}{4\pi\varepsilon_0 r}$$

$$=\frac{\lambda}{4\pi\varepsilon_0}\ln\frac{a+l}{a}$$

$$=2.50\times10^3\ V$$

$$\varphi_b=2\int_0^{\frac{l}{2}}\frac{\lambda\,dr}{4\pi\varepsilon_0\sqrt{r^2+b^2}}$$

$$=\frac{2\lambda}{4\pi\varepsilon_0}\int_0^{\frac{l}{2}}\frac{dr}{\sqrt{r^2+b^2}}$$

$$=\frac{\lambda}{2\pi\varepsilon_0}\ln\left(r+\sqrt{r^2+b^2}\right)\Big|_0^{\frac{l}{2}}$$

$$=\frac{\lambda}{2\pi\varepsilon_0}\cdot\ln\frac{\frac{l}{2}+\sqrt{\frac{l^2}{4}+b^2}}{b}$$

$$=4.30\times10^3\ V$$

13.9 (1)

取任意截面为高斯面，长度为l的圆柱面.

$$\oint\vec{E}\cdot\vec{S}=\frac{q_{in}}{\varepsilon_0}$$

则 $E\cdot2\pi rl=\frac{q_{in}}{\varepsilon_0}$

当 $r\le a$ 时，$q_{in}=\pi r^2 l\rho$

当 $r>a$ 时，$q_{in}=\pi a^2 l\rho$

故 $E=\begin{cases}\frac{r\rho}{2\varepsilon_0} & (r\le a)\\ \frac{a^2\rho}{2\varepsilon_0 r} & (r>a)\end{cases}$

方向沿半径方向

(2) 圆柱内的电势 $(r\le a)$

$$\varphi_{in}=-\int_0^r E\,dr=-\frac{\rho r^2}{4\varepsilon_0}$$

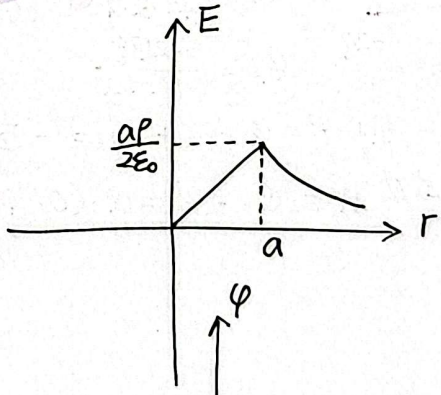
圆柱外的电势 $(r>a)$

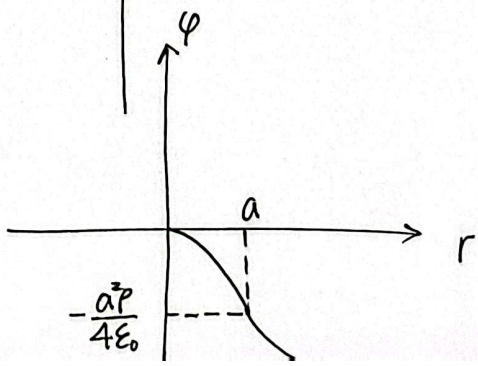
$$\varphi_{out}=-\int_a^r E\,dr-\int_0^a E\,dr$$

$$=-\left(\frac{a^2\rho}{2\varepsilon_0}\ln\frac{r}{a}+\frac{a^2\rho}{4\varepsilon_0}\right)$$

$$=-\frac{a^2\rho}{4\varepsilon_0}\left(2\ln\frac{r}{a}+1\right)$$

(3)





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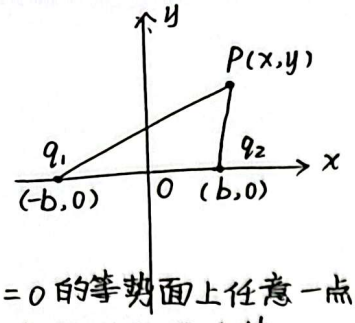
13.16



设 $\varphi=0$ 的等势面上任意一点坐标为 (x,y)

q_1 在 (x,y) 处的电势

$$\varphi_1=\frac{q_1}{4\pi\varepsilon_0 r_1}=\frac{q_1}{4\pi\varepsilon_0\sqrt{(x+b)^2+y^2}}$$

q_2 在 (x,y) 处的电势

$$\varphi_2=\frac{q_2}{4\pi\varepsilon_0 r_2}=\frac{q_2}{4\pi\varepsilon_0\sqrt{(x-b)^2+y^2}}$$

又 $\varphi_1+\varphi_2=0$ 则 $\sqrt{(x-b)^2+y^2}\,q_1=\sqrt{(x+b)^2+y^2}\,q_2$

$$(x-b)^2q_1^2+y^2q_1^2=(x+b)^2q_2^2+y^2q_2^2$$

$$(q_1^2-q_2^2)(x^2+b^2+y^2)=2bx(q_1^2+q_2^2)$$

$$x^2-\frac{q_1^2+q_2^2}{q_1^2-q_2^2}\cdot 2bx+y^2=-b^2$$

$$\left(x-\frac{q_1^2+q_2^2}{q_1^2-q_2^2}b\right)^2+y^2=\left[\left(\frac{q_1^2+q_2^2}{q_1^2-q_2^2}\right)^2-1\right]b^2$$

$$\left(x-\frac{q_1^2+q_2^2}{q_1^2-q_2^2}b\right)^2+y^2=\frac{4q_1^2q_2^2b^2}{(q_1^2-q_2^2)^2}=\left(\frac{2q_1q_2b}{q_1^2-q_2^2}\right)^2$$

可知 $\varphi=0$ 的等势面为球面

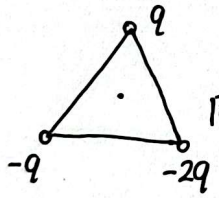
球心为 $\left(\frac{q_1^2+q_2^2}{q_1^2-q_2^2}b,\ 0\right)$，球半径为 $\left|\frac{2q_1q_2b}{q_1^2-q_2^2}\right|$

若两者电量相等，即 $q_1=q_2$

则球心位于无穷远处，球半径为无穷大

即 $\varphi=0$ 的等势面为一个平面，yz平面内

13.20.



$$\varphi_1=\frac{q}{4\pi\varepsilon_0\cdot\frac{1}{\sqrt{3}}a}=\frac{\sqrt{3}q}{4\pi\varepsilon_0 a}$$

同理 $\varphi_2=\frac{-\sqrt{3}q}{4\pi\varepsilon_0 a}$

$$\varphi_3=\frac{-2\sqrt{3}q}{4\pi\varepsilon_0 a}$$

则 $\varphi=\varphi_1+\varphi_2+\varphi_3=-\frac{\sqrt{3}q}{2\pi\varepsilon_0 a}$

外力做功 $W=Q\cdot\Delta\varphi=Q\cdot(0-\varphi)=\frac{\sqrt{3}qQ}{2\pi\varepsilon_0 a}$

扫描全能王 创建

13.22. (1) 电子经过n个圆筒后的速度 v_n

由动能定理有

$\frac{1}{2}mv_n^2 = nqU_0$ 解 $v_n = \sqrt{\frac{2nqU_0}{m}}$

又每次在筒中正好被峰值加速

即每次经过圆筒的时长相等，记为 T_0

则 $v_n \cdot T_0 = L_n$

解 $L_n = \sqrt{\frac{2nqU_0}{m}}\, T_0 \propto n^{\frac{1}{2}}$

又 $L_1 = \sqrt{\frac{2qU_0}{m}}\, T_0$

可知 $L_n = L_1 \cdot \sqrt{n}$

(2) 由题知 $T_0 = \frac{1}{2\nu}$，$q = e$，$m = m_e$

则 $L_1 = \sqrt{\frac{2eU_0}{m_e}} \cdot \frac{1}{2\nu}$

(3) $E_k = neU_0$ (动能定理)

13.29 铀核的静电势能

$$W_U = \frac{3}{5}\frac{Q^2}{4\pi\varepsilon_0 R} = \frac{3}{5} \times \frac{(92 \times 1.6 \times 10^{-19})^2}{4 \times 3.14 \times 8.85 \times 10^{-12} \times 7.4 \times 10^{-15}} = 1.6 \times 10^{-10}\,J$$

两个钯核的总静电势能

$$W = 2W_{Pd} = \frac{6}{5}\frac{Q^2}{4\pi\varepsilon_0 R_1} = \frac{6}{5} \times \frac{(46 \times 1.6 \times 10^{-19})^2}{4 \times 3.14 \times 8.85 \times 10^{-12} \times 7.4 \times 10^{-15} \times \frac{1}{\sqrt[3]{2}}} = \boxed{1.0 \times 10^{-10}}\,J$$

$$= \frac{6}{5}\frac{\frac{Q^2}{4}}{4\pi\varepsilon_0 \frac{R}{\sqrt[3]{2}}} = \frac{2\sqrt[3]{2}}{4}\frac{Q^2}{4\pi\varepsilon_0 R} = \frac{\sqrt[3]{2}}{2} W_U$$

裂变释放的静电势能

$\Delta W = W_U - W = 6 \times 10^{-11}\,J$

1kg铀裂变释放的静电能

$\Delta W' = \frac{1000}{235} \cdot \Delta W = 1.5 \times 10^{14}\,J$

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